

Chapter 11

Titration in Analytical Chemistry

Fundamentals of Analytical Chemistry, Skoog, et al., 10th Ed., 2022

Chapter 11 Problems:

11-3, 11-7, 11-15, 11-19, 11-21, 11-23, 11-27a

Standard Solution (aka standard titrant): reagent of known concentration

Back-Titration: process in which the excess standard titrant used to consume an analyte is determined by titration with a second standard solution. Often used with the rate of reaction is slow or when the standard solution lacks stability.

Equivalence Point is the theoretical point in a titration when the exact stoichiometric amount of titrant has been added is to fully react with the analyte

End point is the point in a titration when a physical change occurs that is associated with the condition of chemical equivalence

Titration Error is the difference in volume between the equivalence point and the end point.

$$E_t = V_{ep} - V_{eq}$$

Indicators are often added to an analyte solution to produce an observable change (signaling the end point)

Primary standard is an ultrapure compound that serves as the reference material for a titration or for another type of analysis

1. High Purity
2. Atmospheric stability
3. Absence of hydrate water (so it doesn't change with changing humidity)
4. Modest Cost
5. Reasonably soluble
6. Reasonably large MM so that the relative error associated with weighing it is minimized.

Standard Solutions

- 1. Be sufficiently stable so that it necessary to determine its concentration only once**
- 2. React rapidly with the analyte so that the time required between additions of reagent is minimized**
- 3. React more or less completely with the analyte so that satisfactory end points are realized**
- 4. Undergo a selective reaction with the analyte that can be described by a balanced equation**

The accuracy of a titration can be no better than the accuracy of the concentration of the standard solution used.

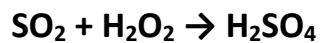
Two methods for determining the concentration of a standard solution

- 1. Direct method: Carefully determined mass of primary standard is dissolved in a solvent in a volumetric flask.**
- 2. Standardization: a known mass of primary standard is titrated with the solution to be standardized
(a titrant that is standardized is a secondary standard solution)**

1. A 25.00 mL aliquot of HNO_3 required 27.21 mL of 0.01479 M Na_2CO_3 solution to reach the endpoint. Calculate the molarity of the nitric acid.

2. A 0.2178g sample of impure magnesium hydroxide was dissolved in 50.00 mL of 0.1204 M HCL. Back-titration of the excess acid required 3.76 mL of 0.0948 M NaOH. Calculate the percentage of $\text{Mg}(\text{OH})_2$ in the sample.

3. The sulfur in an organic compound was determined by burning 0.471 g sample in a stream of oxygen and collecting the resultant SO_2 in a neutral solution of H_2O_2 , which converted the SO_2 to H_2SO_4 :



The sulfuric acid was titrated with 28.20 mL of 0.1080 M KOH. Calculate the % sulfur in the sample.